Project Proposal

Active Feature-Value Acquisition with Sampled Expected Utility

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1. Background and Motivation

Real-world machine learning data often contain missing feature values, and in many domains each missing value can be acquired at a non-trivial cost. In this setting, the bottleneck is not only label acquisition but also *which feature values* should be acquired first to improve model quality most efficiently. In many practical pipelines, collecting complete information for all instances is unnecessary and expensive. Instead, we can start from partially observed data and actively acquire only those missing values that are most useful for the downstream classifier.

Our project aims to implement *Active Feature-Value Acquisition* (AFA), following the framework introduced by Saar *et. al* [1]. We focus on the scenario where class labels are known, but some feature entries are missing and costly to obtain. The objective is to iteratively select feature-value queries that maximize expected improvement in predictive performance per unit cost.

Compared with passive acquisition (uniform random acquisition), AFA is expected to reach a target accuracy with lower total acquisition cost. This is especially relevant when features are heterogeneous in informativeness and when acquisition resources are limited.

2. SEU Strategy

We adopt the SEU (Sampled Expected Utility) strategy for active feature-value acquisition. Let $F_{i,j}$ denote a missing value (instance i, feature j), and $q_{i,j}$ the corresponding acquisition query. For each candidate query, SEU computes an expected utility score:

$$\mathrm{E}(q_{i,j}) = \sum_{k=1}^{K} U(F_{i,j} = v_k)\, P(F_{i,j} = v_k),$$

where v_k are possible values of feature j, $P(F_{i,j} = v_k)$ is the estimated probability of each value (conditioned on currently observed features for instance i), and $U(\cdot)$ is the utility contribution normalized by acquisition cost. Utility is computed as:

$$U(F_{i,j} = v_k) = \frac{\Delta(F \cup \{F_{i,j} = v_k\})}{C_{i,j}},$$

where $\Delta(\cdot)$ denotes the model performance gain (measured on a held-out validation set) after acquiring the value, and $C_{i,j}$ is the acquisition cost.

We will use *log-gain* as the utility-sensitive performance signal instead of raw classification accuracy, because log-gain better captures confidence changes that may later reduce classification error.

The overall acquisition loop proceeds as follows. Given partially observed data, we first train a Naive Bayes classifier on the available values, where Naive Bayes serves both as the base classifier and as the probability estimator for $P(F_{i,j} = v_k)$. At each round, SEU scores are computed for all missing entries using the current

model; the highest-scoring feature value is then acquired from the oracle, and the classifier is retrained on the updated dataset. This process repeats until the acquisition budget is exhausted.

To reduce computation, SEU evaluates only a sampled subset of candidate queries at each round. We will implement two sampling variants:

- **SEU-US:** reduce candidates with uniform random query sampling.
- **SEU-ES:** reduce candidates with error sampling, prioritizing misclassified or uncertain instances.
- **SEU-Accuracy (ablation):** replace log-gain with classification accuracy as utility.

3. Proposed Aims

The project has two main aims.

Aim 1: Implement SEU for known labels.

We will use the UCI Breast Cancer dataset with categorical features. To simulate partial observability, we will randomly mask feature values at two missingness levels (10% and 20%). Given known labels, we apply the SEU acquisition loop described in section 2 to iteratively acquire missing feature values and retrain the classifier. In this aim we focus on the core SEU scoring loop; the two sampling variants (SEU-US and SEU-ES) are evaluated together in Aim 2.

Aim 2: Compare SEU variants and baselines under equal acquisition budget.

We will compare the following methods:

- **Uniform baseline:** random acquisition of missing feature values.
- **SEU-US:** sampled expected utility with uniform candidate reduction.
- **SEU-ES:** sampled expected utility with error-sampling candidate reduction.
- **SEU-Accuracy:** expected utility using accuracy instead of log-gain.

As a reference ceiling, we also train a classifier on fully observed data (all true feature values known); this upper-bound model represents the best attainable performance under complete information.

Performance will be measured by accuracy/error reduction versus cumulative acquisition cost. The main question is whether SEU, especially SEU-ES, achieves better model quality for the same cost budget.

4. Predicted Outcomes

We expect the full simulation pipeline to be computationally feasible with repeated random masks and multiple rounds of acquisition. If the method behaves as reported in prior work, SEU should outperform uniform random acquisition, especially at earlier acquisition stages. In particular, we expect:

- **SEU-ES** will yield the largest average per-round error reduction among tested variants.
- **SEU** methods will reach target performance with lower total acquisition cost than uniform sampling.
- Using **log-gain** will outperform the **SEU-Accuracy** ablation, validating utility sensitivity.
- Improvements will remain observable under both 10% and 20% missingness settings.
- The gap between acquisition-based methods and the fully observed upper bound will quantify remaining headroom.

5. Timeline

Week 1: Prepare the dataset and build the masking pipeline (10% and 20% random missingness).

Week 2–3: Implement core SEU scoring, model retraining loop, and batch acquisition.

Week 4: Implement comparison variants and run main experiments.

Week 5: Analyze results and complete the final report.

Challenges The main challenges are:

- **Probability estimation under sparse observations:** estimating $P(F_{i,j} = v_k)$ can be noisy with limited known values.
- **Computation cost of scoring:** expected utility requires repeated model updates and can be expensive without candidate reduction.
- **Sensitivity to masking randomness:** results may vary with different missing-value patterns and seeds.

These risks will be mitigated by repeated runs with fixed seeds, reporting mean and variance across trials, and including both SEU-US and SEU-ES reductions.

6. Resources

Data source: UCI Breast Cancer dataset. Software stack: Python, pandas, numpy, scikit-learn, and standard plotting libraries. Reference method: Active Feature-Value Acquisition [1] and ASR lecture notes (Lecture 13, slides 56–76).

References

[1] Maytal Saar-Tsechansky, Prem Melville, and Foster Provost. "Active Feature-Value Acquisition". In: *Management Science* 55.4 (2009), pp. 664–684.